Case 1: Premium-certified helmets have 2.5x the head-injury rate

First objection:

The prescribed zero-baseline bars reproduce 2.15 versus 5.40; the problem is not a truncated axis.

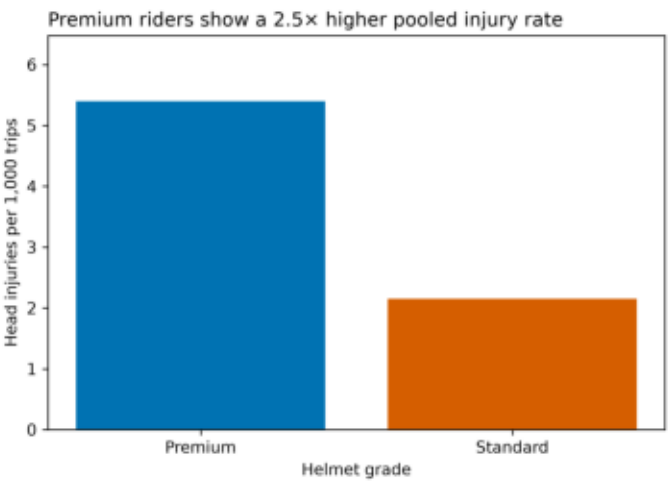
Mechanism and data evidence:

Road-type exposure confounds the pooled rates: Premium has 34,000/40,000 motorway trips versus Standard 2,000/60,000. Within city, inner-city and motorway, Premium rates (2.20, 1.00, 6.00) are below Standard (3.00, 1.39, 8.00).

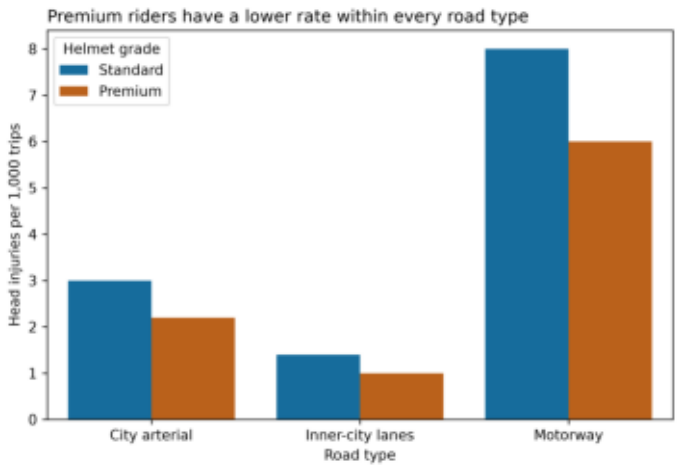
Redesign:

Stratify by road type; pooled rates do not show helmet harm.

Rebuild



Redesign



Case 2: Canteen chai sales predict quiz failures

First objection:

The fitted line is faithful to the recipe and r=0.982; the issue is not a plotting error.

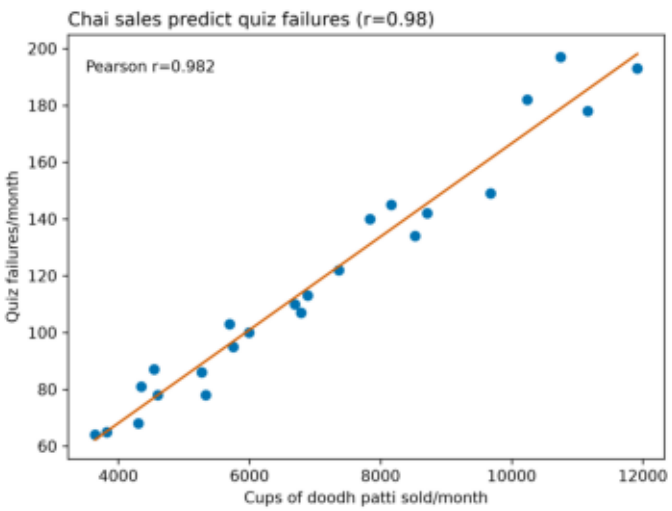
Mechanism and data evidence:

Exposure/time trend: cups correlate with enrollment r=0.997 and failures with enrollment r=0.977; normalized cups/rate r=0.293, while failure rate tracks exam month r=0.879.

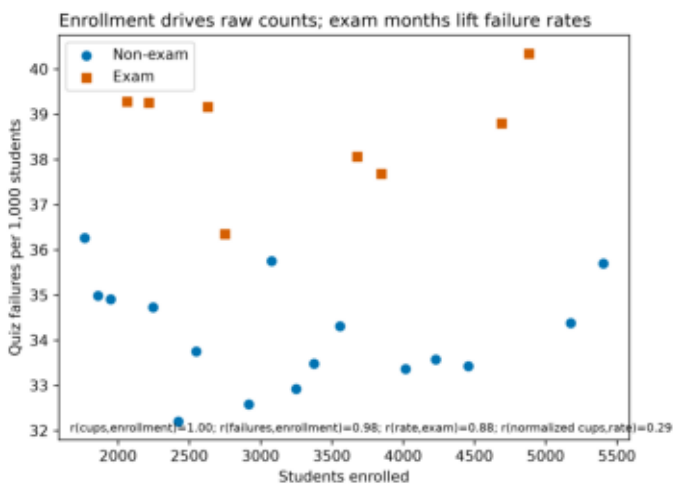
Redesign:

Compare exposure-normalized failure rates by enrollment and exam month; counts alone do not imply chai causes failures.

Rebuild



Redesign



Case 3: Smog Action Plan cut Lahore AQI by a third

First objection:

Zero baseline is honored; means are 202 versus 137 (−32%).

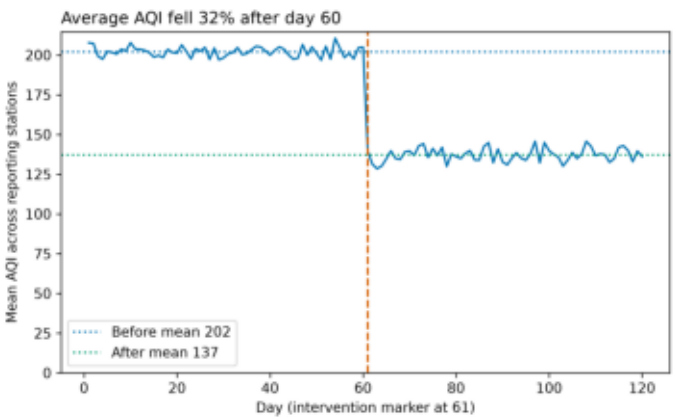
Mechanism and data evidence:

Coverage falls from 22 to 13 reporting stations on day 61; Industrial readings fall 360→0. Their pre-period mean is 322, so composition changes when the most polluted zone disappears.

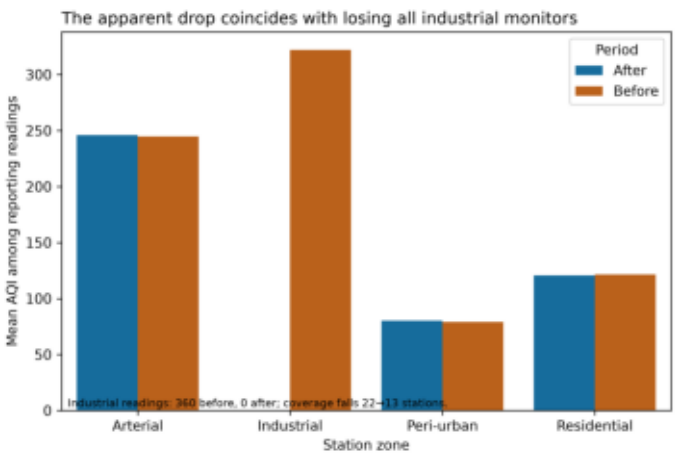
Redesign:

Show zone means and a fixed-station panel (n=13 complete stations); the network cannot establish a policy effect after industrial coverage vanishes.

Rebuild



Redesign



Case 4: Four departments behave identically, so fund them equally

First objection:

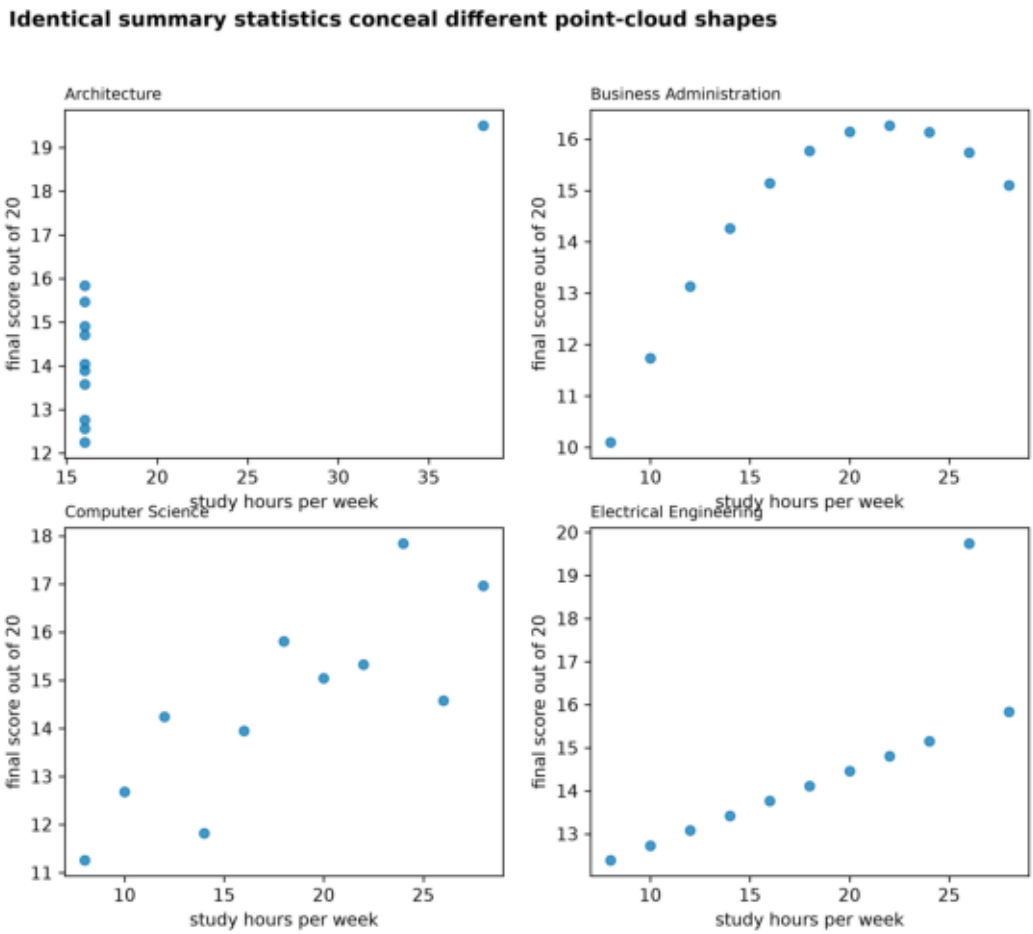
The repeated moments are deliberately constructed and correctly calculated; this is not a rounding or sample-size mistake.

Mechanism and data evidence:

Equal n=11, means 18/14.5 and r≈.816 do not imply identical distributions; cohort assignments and joint shapes differ.

Redesign:

Keep the required summary table and inspect the labeled small-multiple scatterplots before funding or designing interventions.



Case 5: Five cohorts, one set of statistics

First objection:

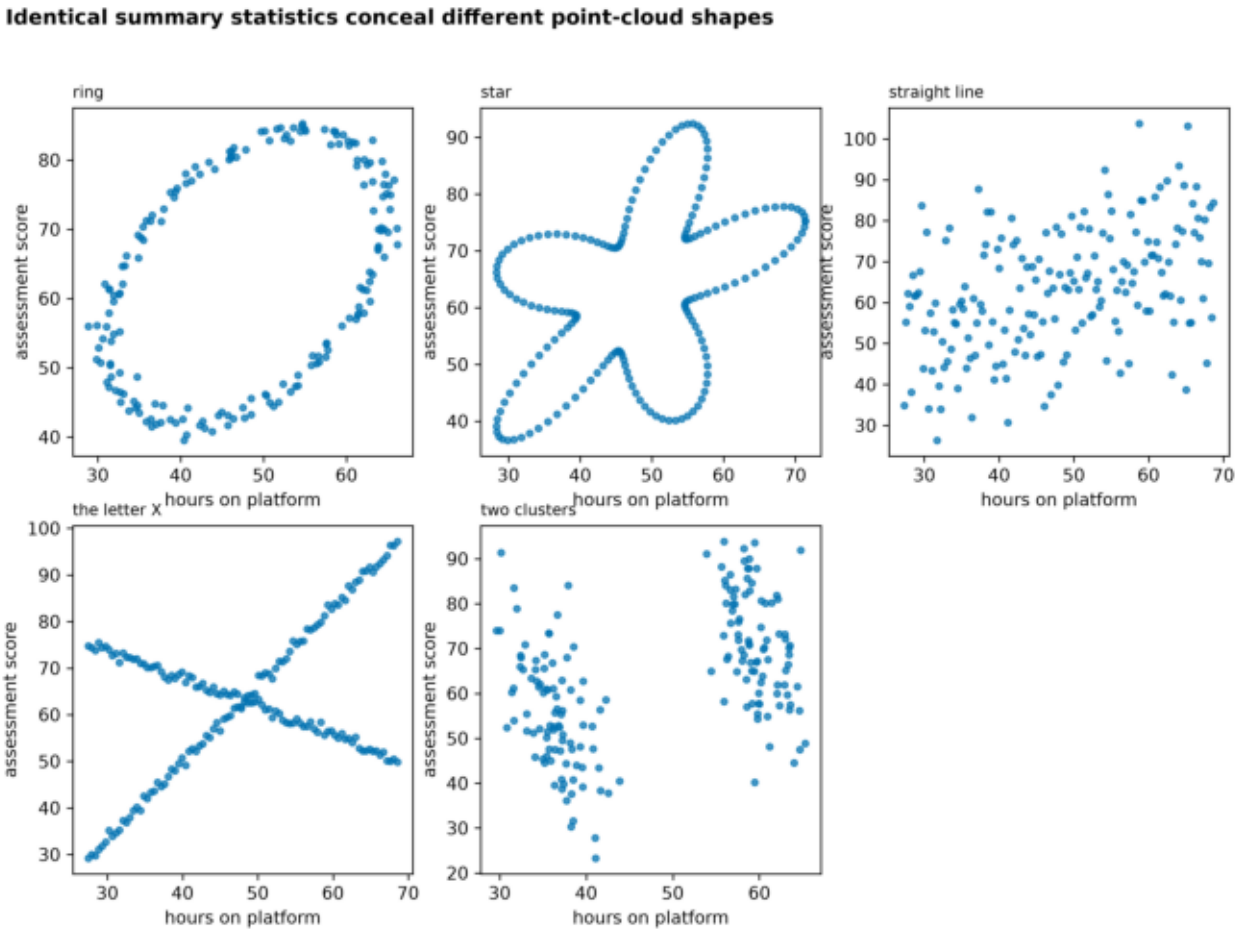
The repeated moments are deliberately constructed and correctly calculated; this is not a rounding or sample-size mistake.

Mechanism and data evidence:

Each n=180 group has mean 48/63, SD 12/15 and r=.420, yet shapes include line, ring, star, X and clusters; moments discard geometry.

Redesign:

Keep the required summary table and inspect the labeled small-multiple scatterplots before funding or designing interventions.



Case 6: Creatinine tracks HbA1c — a finding

First objection:

The matrix is 20×20 and the pair is r=.765; sample size or a wrong pair is not the issue.

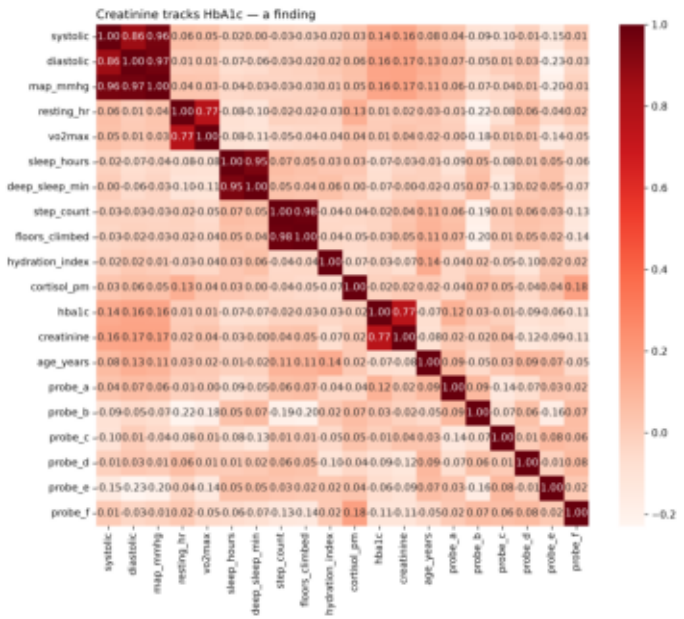
Mechanism and data evidence:

Two defects: sequential Reds has no neutral midpoint for signed r, and the mirrored triangle repeats 190 pairs. The data does not identify a causal driver: age correlations are −.076/−.068. Systolic–MAP r=.96 is partly arithmetic because MAP=(systolic+2×diastolic)/3.

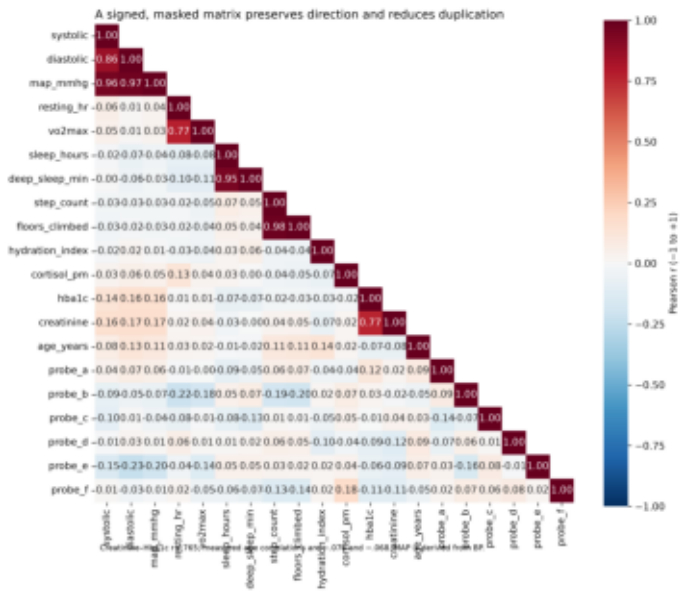
Redesign:

Use a diverging map fixed −1…+1 and mask one triangle; inspect the target pair, but do not claim a clinical mechanism from correlation.

Rebuild



Redesign



All case-study datasets are synthetic for CDB601220; no real patient or rider records.

Case 7: ShaheenGo ran at a loss until May

First objection:

Both series share PKR millions, so separate scales have no unit justification.

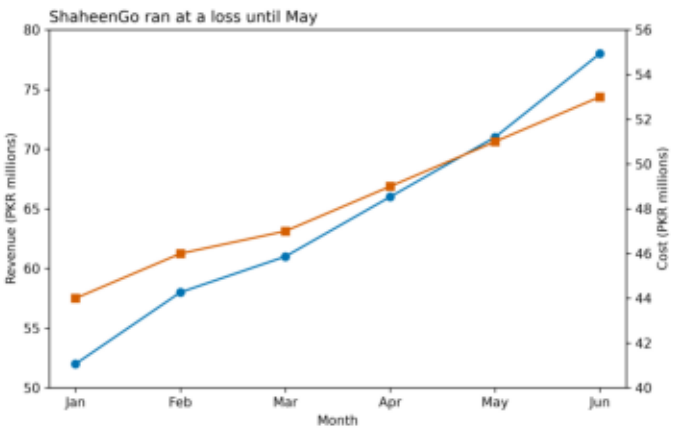
Mechanism and data evidence:

Independent y-axis transformations make visual vertical positions incomparable. Actual margins are +8,+12,+14,+17,+20,+25 PKR million; revenue exceeds cost in every month.

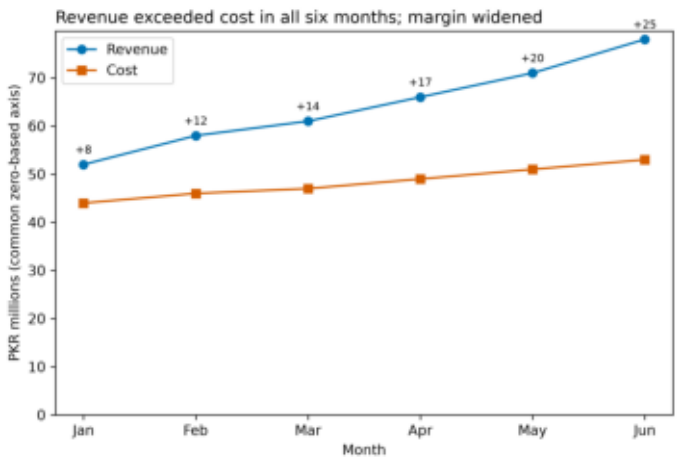
Redesign:

Use a common zero-based axis and direct margin annotations; never use twinx for two comparable measures.

Rebuild



Redesign



Case 8: Three bin widths, three different distributions

First objection:

Every panel uses the same n=10,000 values; sample size is fixed.

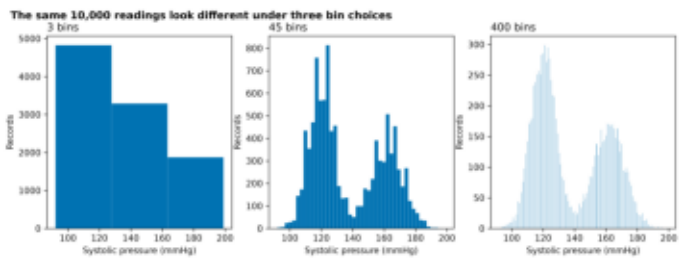
Mechanism and data evidence:

Binning is a display parameter: 3 bins hide detail, 400 fragment counts into noise, and the middle choice is still arbitrary.

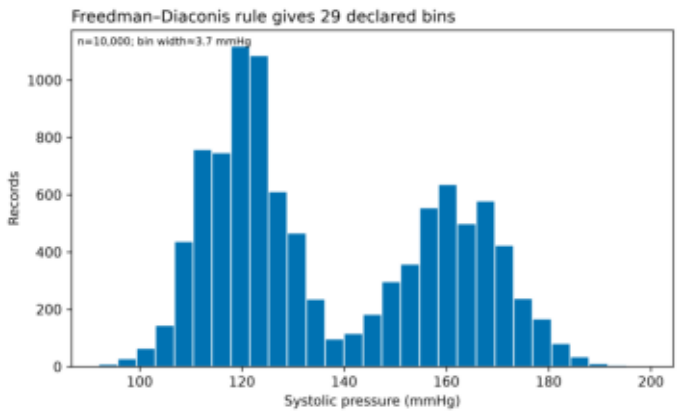
Redesign:

Use the explicit Freedman–Diaconis rule (29 bins, mean width 3.7 mmHg) and declare it.

Rebuild



Redesign



Case 9: 4.4% of the density sits above 100% saturation

First objection:

Observed maximum is exactly 100; no record exceeds the physical ceiling.

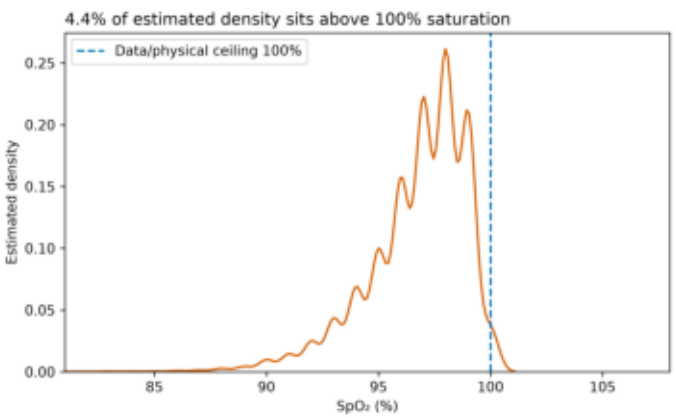
Mechanism and data evidence:

Gaussian KDE convolves discrete bounded observations with symmetric kernels; default bandwidth and cut=3 extend density past the maximum.

Redesign:

Use the empirical integer distribution or a bounded estimator; report the 100% ceiling.

Rebuild



Redesign

